

Multiple Choice (10 marks)

1. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
2. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web
3. Why would a ping sweep be used?
 - (a) To identify live systems
 - (b) To locate live systems
 - (c) To identify open ports
 - (d) To locate firewalls
4. Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is NOT complete?
 - (a) {AND, NOT}
 - (b) {NOT, OR}
 - (c) {AND, OR}
 - (d) {NAND}
5. Of the following sorting algorithms, which has a running time that is LEAST dependent on the initial ordering of the input?
 - (a) Insertion sort
 - (b) Quicksort
 - (c) Merge sort
 - (d) Selection sort

True / False (10 marks)

Answer True or False

6. True or False: Nmap categorizes port states into active, inactive, and standby based on the network conditions.
7. True or False: A buffer-overflow attack involves exploiting memory vulnerabilities by injecting extra data with malicious instructions.
8. True or False: The Opera browser is specifically designed for accessing the Tor network and its services.
9. True or False: Escaping queries is an effective method to prevent SQL injection attacks by neutralizing malicious input.
10. True or False: The regular expression $(a + b)^*c + (a + b)^*d$ is equivalent to $(a^* + b)^*(c + d)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the equilibrium index of the given array.
12. Write a python function to find the sum of squares of first n even natural numbers.

End of Paper